



INKSPECT

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Solving Global Health

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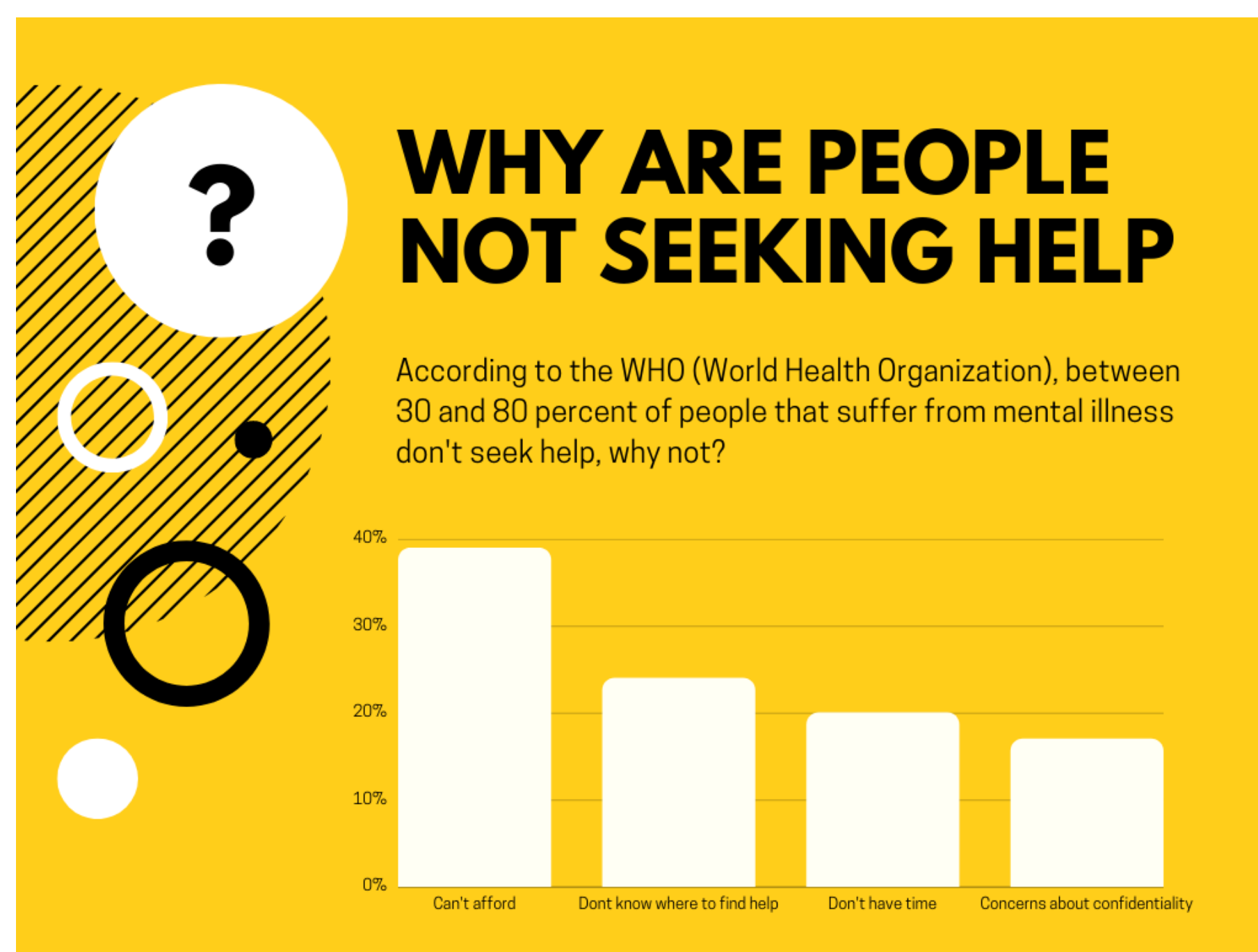
Abstract

A great proportion of people who suffer mental illnesses do not seek help. To solve this problem, we present a design and implementation plan for a more accessible psychology test: Inkspect. Based on the web, Inkspect will acquire user inputs for the Rorschach inkblot responses and use Natural Language Processing (NLP) to score them based on the Rorschach Comprehensive System (RCS). Then the RCS will assess the person's psychological state. Inkspect is expected to promote global mental health.

Problem Statement

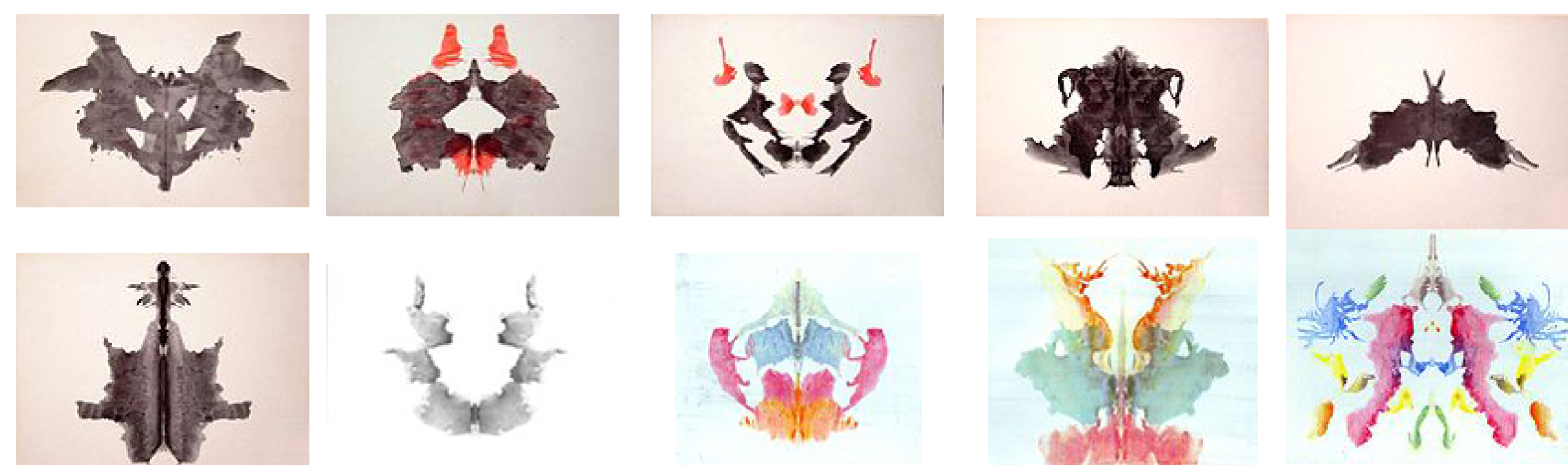
According to the World Health Organization (WHO), almost one billion people had mental disorders in 2019. Exacerbated by the Covid-19 pandemic and the 2022 Russian invasion of Ukraine, these numbers have increased significantly [1]. For adolescents aged 12-17, 31% suffer illnesses such as depression, anxiety, or ADD/ADHD [2].

Despite the seriousness of the current state, many do not seek help. Accessibility is limited and people are not willing to spend money and time to see a psychiatrist. To add, concerns on confidentiality deter them from taking actions.



Rorschach Test

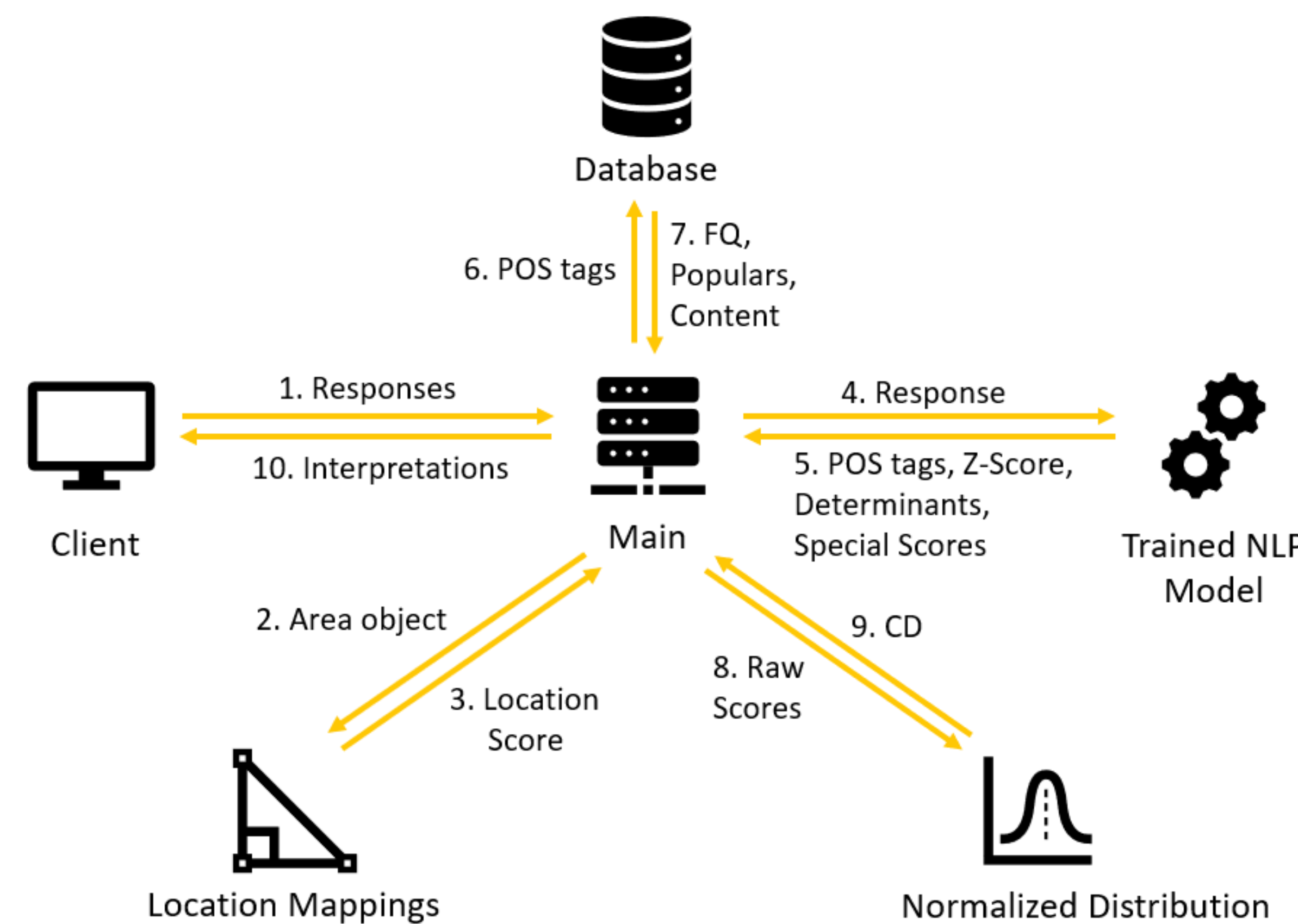
Rorschach's test, also known as the inkblot test, is a projective psychological test that analyzes the subjects' perceptions of 10 ambiguous inkblots: 5 in black and white and 5 in color.



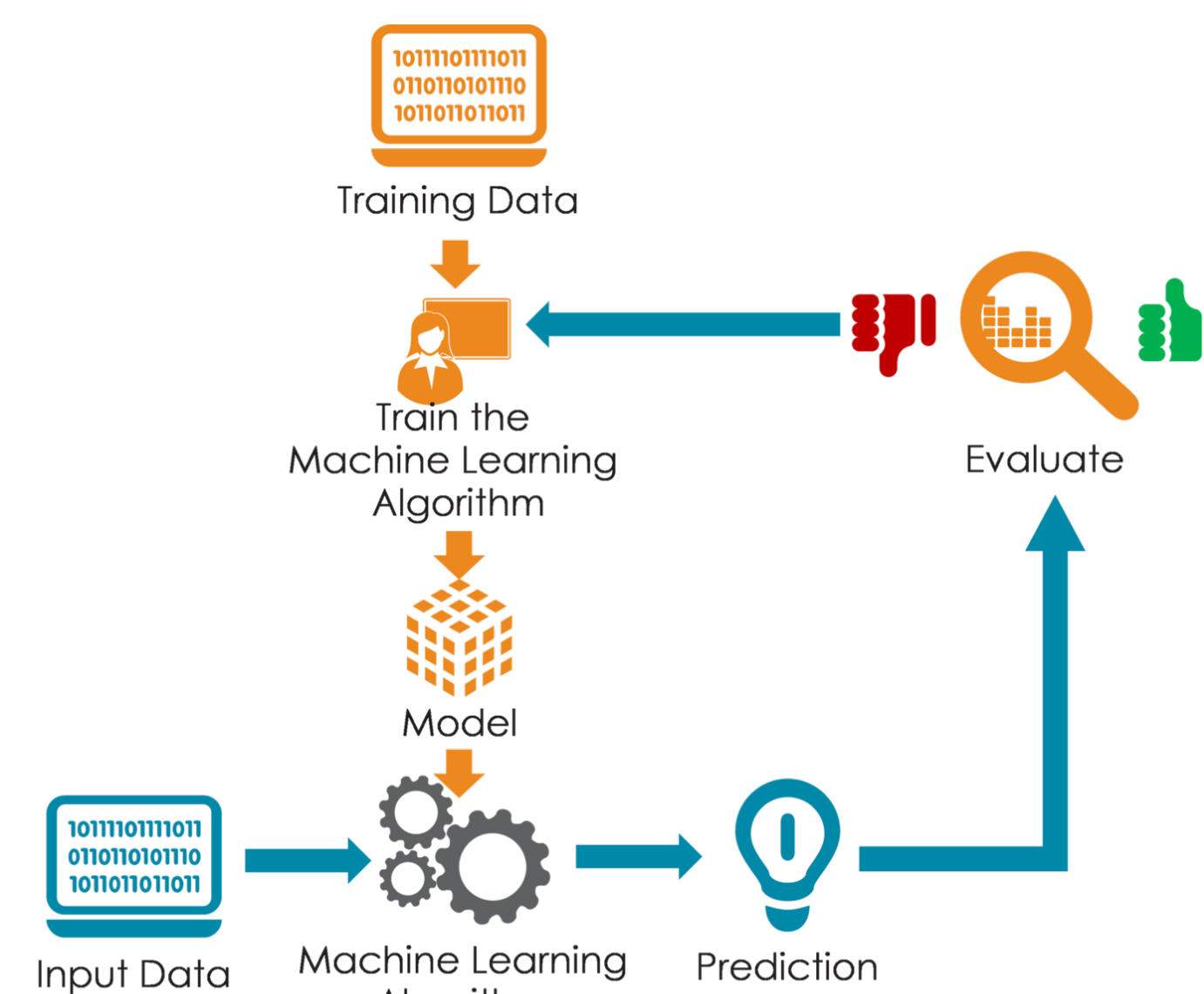
The test consists of two phases: the response phase and the inquiry phase. During the response phase, the subject is asked *what* the blots look like. In the inquiry phase, the subject is asked *where* and *why* they see the image. The responses are scored according to the RCS, then interpreted. RCS is an extensively validated model that achieves a high inter-rater reliability which makes it viable for automation [3].

During our research, we have concluded that no online Rorschach test performs the inquiry phase successfully. The use of multiple-choice questions in the inquiry phase suggests serious misinterpretations, while some websites did not even have an inquiry phase. Moreover, most of these websites contained ads which distracted from purpose of psychological analysis and leaned towards monetization purposes.

Software Design



Our team aims to create Inkspect, a website that will inspect inkblot tests. A key difference from the other tests is that it will follow the administrative procedures of RCS (though it will have certain limitations such as gesture detection) with sophisticated analysis [4]. While the *what* and *why* information is obtained via textboxes, the *where* information will be obtained by creating a lasso tool to select an area. Inkspect will use automated replies in order to guide the user through the test. The Responses will be sent to the back-end server where it will go through filters for each scoring category. Such scoring categories are location, developmental quality (DQ), determinants, form quality (FQ), contents, popular responses, Z-scores, and special scores. Using Part-of-Speech Tagging (POS) on responses obtained during the response phase, keywords can be extracted and can be queried to a database acquired through the CHESSSS open source software [5]. The database will provide scores for content, FQ, and popular responses. The responses obtained during the inquiry phase will go through NLP semantic analysis to identify determinants and special scores, while co-reference resolution can be used to score for DQ and Z-scores. After the test is fully administered, the results will be compared with normalized results using cumulative distribution (CD) and displayed in a spectrum in the order of significant psychological traits [4]. As there is no written result produced, the user determines whether he/she should see a psychiatrist for the full diagnostics and possible prescriptions. Overall, the project is feasible.



The NLP model will be trained using data obtained from open source datasets and lexical databases then evaluated based on training datasets obtained from the Internet as well as from Rorschach test practitioners.

Project Management

Task	June 25 - July 5	July 6 - July 15	July 16 - July 25	July 26 - August 5
Research & Planning				
Obtain Professional Opinion (1)				
Data Acquisition				
Rapid Development				
Further Development				
Deployment				
Documentation				

Our team has successfully delegated tasks which increased the efficiency and overall quality of the project. After the working prototype is produced, the project should get more professional opinion before continuing any further.

Impacts

Our overall aim is to try and help as many people as possible through a simple and accessible test that can yield reasonable and accurate results. Not limited to cost and time, Inkspect ensures confidentiality by not requiring the user to enter his/her name and storing none of the data provided by the user. When provided with honest and unambiguous responses, Inkspect will assist users to move forward and get professional help.

Mental Disorders	Average Non-treatment Rates (%)
Schizophrenia	32
Bipolar Disorder	50
Panic Disorder	55
Major Depression	56
Generalized Anxiety Disorder	57
Obsessive-Compulsive Disorder	59
Alcohol Dependence	78

Table 1. The average non-treatment rates for specific disorders. The rates are acknowledged to be lower than the actual numbers. [6]

In 2014, 62.7% of people with mental illnesses in the UK did not seek for mental health treatment [7]. As shown in Table 1, many people require mental health treatment. These statistics manifests the potential impact Inkspect can have on the global mental health environment.

References

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